

# Development history: filtering, the C-score, and LD clustering

The full record, including what was tried and withdrawn

module\_sim\_LDscnR

01 September 2026

## Contents

|                                                                              |           |
|------------------------------------------------------------------------------|-----------|
| <b>1 Purpose of this document</b>                                            | <b>2</b>  |
| <b>2 Scope, and what changed after review</b>                                | <b>2</b>  |
| 2.1 Terminology . . . . .                                                    | 2         |
| <b>3 Why the ranking approach was abandoned</b>                              | <b>2</b>  |
| <b>4 Design</b>                                                              | <b>3</b>  |
| 4.1 What is compared . . . . .                                               | 3         |
| 4.2 Composition of the panels . . . . .                                      | 3         |
| 4.3 The multiplicity mechanism, and its cost . . . . .                       | 3         |
| <b>5 Results, pooled — with ties and magnitudes</b>                          | <b>5</b>  |
| <b>6 Stratified results</b>                                                  | <b>5</b>  |
| <b>7 Null behaviour: two nulls, and they disagree</b>                        | <b>8</b>  |
| 7.1 Which null is the right one . . . . .                                    | 9         |
| 7.2 What this does and does not establish . . . . .                          | 9         |
| <b>8 Status of each claim</b>                                                | <b>9</b>  |
| <b>9 Relationship to the empirical panel analysis</b>                        | <b>10</b> |
| 9.1 What EMMAX is actually run on . . . . .                                  | 10        |
| 9.2 What the representative costs, measured at locus level . . . . .         | 10        |
| 9.3 Why the single-SNP threshold is the looser one . . . . .                 | 10        |
| 9.4 How a cluster should be summarised . . . . .                             | 12        |
| 9.5 The residual cost, and its mechanism . . . . .                           | 12        |
| 9.6 Relationship to the empirical panel analysis . . . . .                   | 12        |
| <b>10 Corrections applied after review</b>                                   | <b>13</b> |
| 10.1 Open, and in progress . . . . .                                         | 13        |
| <b>11 Geometric exclusion of long-and-sparse clusters: a negative result</b> | <b>13</b> |
| 11.1 Why the inference was wrong . . . . .                                   | 14        |
| 11.2 What the accumulated negatives now show . . . . .                       | 14        |
| <b>12 Three deferred analyses, reported</b>                                  | <b>14</b> |
| 12.1 Does the result survive the newer decay settings? . . . . .             | 14        |
| 12.2 Where does the $k$ curve turn? . . . . .                                | 15        |
| 12.3 Do the surrogate nulls exercise structure? . . . . .                    | 15        |

|                                                                        |    |
|------------------------------------------------------------------------|----|
| 13 Withdrawn: the confounding mechanism relayed from the panel session | 15 |
| 14 Provenance                                                          | 16 |

## 1 Purpose of this document

This is the **working record**. It keeps every line of enquiry, including those that failed, every claim that was withdrawn, and the reason each correction was made. It exists so that a result cannot quietly revert to a superseded version and so that the reasoning behind the final method is auditable.

A companion document, `ld_clustering_result.pdf`, states only what survived and is written for a reader who does not need this history.

**Where this ended up.** Two things were tried on the same premise — that  $ld_w$ , being phenotype-blind, carries information about where causal variation sits.

**Using  $ld_w$  to rank markers (the C-score): abandoned.** It does not beat BH at any deployable threshold.

**Using  $ld_w$  to select which units to test ( $k$ -filtering): abandoned.** It buys recall at a steep precision cost, requires a free parameter  $k$  with no principled value, and was outperformed by doing nothing.

**Using  $ld_w$  to define the units (LD clustering): kept.** This is the result in the companion document. It has no free parameter, and its central argument — that BH over near-independent clusters controls FDR over loci rather than over SNPs — requires no knowledge of causal positions.

## 2 Scope, and what changed after review

This describes a **selection** analysis: choosing which units to test before the phenotype is seen. It shares a premise with the abandoned C-score work —  $ld_w$  is phenotype-blind and may carry information about where causal variation sits — but differs in unit, truth, comparison and conclusion.

An earlier draft was reviewed externally. That review found two factual errors, an ambiguous denominator, and — most consequentially — that pooling 80 panels across four parameter cells concealed the main structure in the data. **This version is stratified, and the conclusion changed as a result.** All corrections are recorded in Section~10 rather than silently applied.

**The finding, as the evidence now supports it.** Pre-filtering LD clusters increases QTN-proximal discoveries at moderate-to-large filter sizes — by a median of about **one** such discovery per panel, against four to five extra total calls. Which filter is better **depends on the regime**: local-LD weight outperforms cluster size where background LD is low, and fails where it is high. Neither filter is established as generally preferable.

### 2.1 Terminology

The endpoint is **not** a true discovery. Truth here is bp proximity to a driving QTN with the causal variants themselves excluded, so a cluster at 49 kb counts and one at 51 kb does not, regardless of actual LD. Throughout this document such discoveries are called **QTN-proximal**. “True discovery” is reserved for a rule tied to the simulated causal architecture, which this is not.

## 3 Why the ranking approach was abandoned

Recorded for self-containment. All paired within run with a sign test.

| Test                                 | Result                                              |
|--------------------------------------|-----------------------------------------------------|
| $\tau_C = 0.05$ vs $\alpha = 0.05$   | C better in 79/151 non-tied (52%), $p = 0.63$       |
| PR-AUC at the operative distance cap | C better in 120/313 (38%), $p = 4.4 \times 10^{-5}$ |
| Stage-2 clusters as fixed units      | $\alpha$ ahead by 0.07 PR, flat at every size floor |

Two external results agree: the 3sp panel session finds C *ties* an oracle best-single-cell, and the simulation-parsing session measured the mechanism — reweighting a p-value by  $ld_w$  nets about **one marker in 1,400** at 50–100 kb and nothing at 10 kb.

## 4 Design

### 4.1 What is compared

| Component | Choice                                                                           | Why                                                                                                                                                      |
|-----------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unit      | Stage-2 LD cluster, tested through its LD-central pruned representative          | Representative = highest median $r^2$ to the other members. Membership recomputed per bundle and required to reproduce the stored GRM marker set exactly |
| Filter    | Top $k$ units by median $ld_w$ ; computed before any phenotype                   | Compared against MAF, cluster size and random at identical $k$                                                                                           |
| Test      | BH at 0.05 within the selected set only                                          | Identical to the conventional analysis, applied to fewer tests                                                                                           |
| Endpoint  | QTN-proximal: within 10 / 50 / 100 kb of a driving QTN, causal variants excluded | Shares no machinery with $ld_w$ ; an $r^2$ -tagging truth would flatter it by construction                                                               |
| Baseline  | BH at 0.05 over all units                                                        | This IS the conventional $\alpha$ analysis, so the comparison is built in                                                                                |

### 4.2 Composition of the panels

The 80 panels are 4 parameter cells  $\times$  2 background-selection arms  $\times$  10 environments, each pooling 10 map sets into a genome-wide panel of about 86,000 stage-2 clusters. Cells and arms are **not** interchangeable and are stratified throughout Section~6.

### 4.3 The multiplicity mechanism, and its cost

Filtering changes no p-value. It changes the BH threshold, because BH over 99,786 units is far more severe than BH over 5,000. On the illustrative panel below the accounting is **27 discoveries – 1 lost + 28 gained = 54**: filtering can and does *remove* an already-significant result when it falls outside the selected set. Of the 28 gained, **6 (21%) are QTN-proximal**, so precision on gained units is low by construction.

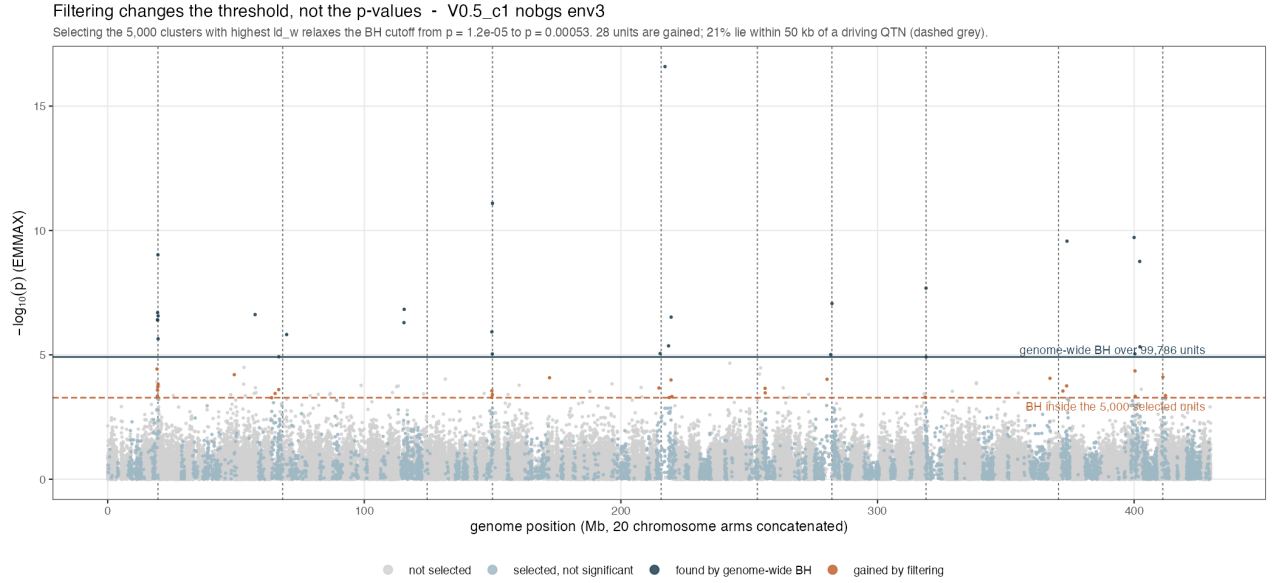


Figure 1: The mechanism. Navy: cleared the genome-wide cutoff. Orange: lie between the two cutoffs, gained by filtering alone. Grey dashed verticals mark driving QTN. One navy unit outside the selected set is lost and is not shown separately.

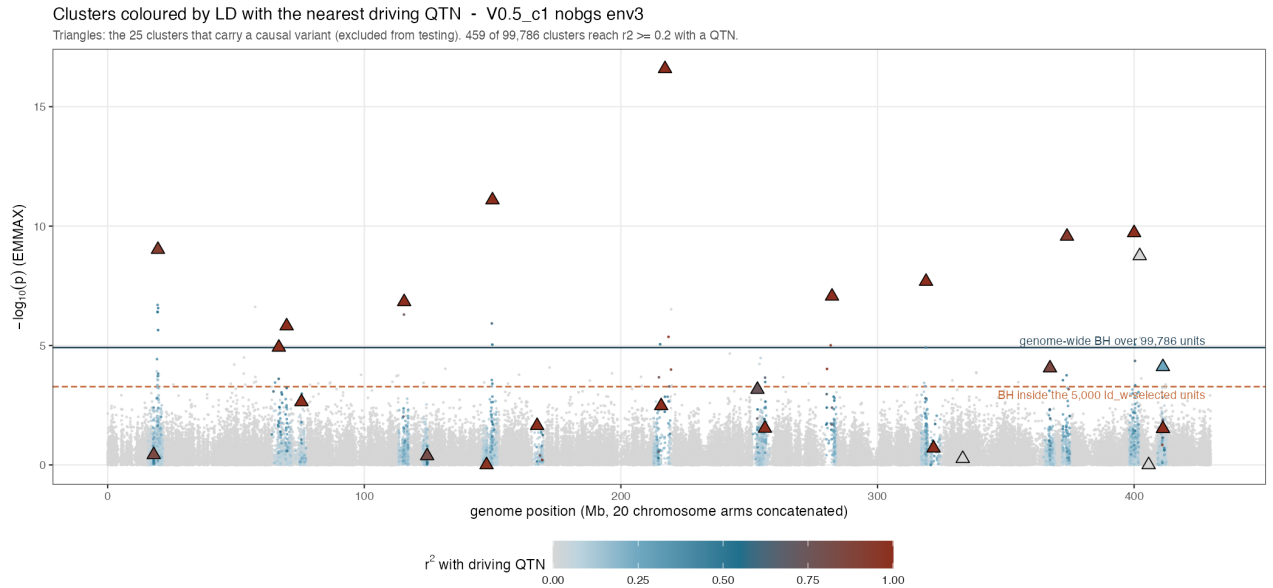


Figure 2: Units coloured by  $r^2$  with the nearest driving QTN; triangles are the 25 clusters carrying a causal variant, which are excluded from testing. Only 459 of 99,786 clusters reach  $r^2 \geq 0.2$  with any QTN, and several QTN-carrying clusters sit at  $p \approx 1$ .

## 5 Results, pooled — with ties and magnitudes

| Filter | $k$    | wins | losses | ties | % non-tied | median | IQR     | $p$   |
|--------|--------|------|--------|------|------------|--------|---------|-------|
| ld_w   | 500    | 33   | 24     | 23   | 58         | +0.0   | [-2, 2] | 0.289 |
| ld_w   | 1,000  | 41   | 24     | 15   | 63         | +1.0   | [-1, 2] | 0.046 |
| ld_w   | 2,000  | 46   | 17     | 17   | 73         | +1.0   | [0, 3]  | 3e-04 |
| ld_w   | 5,000  | 52   | 14     | 14   | 79         | +1.0   | [0, 3]  | 3e-06 |
| ld_w   | 10,000 | 50   | 12     | 18   | 81         | +1.0   | [0, 2]  | 1e-06 |
| ld_w   | 20,000 | 47   | 0      | 33   | 100        | +1.0   | [0, 2]  | 1e-14 |
| size   | 500    | 50   | 17     | 13   | 75         | +1.0   | [0, 3]  | 7e-05 |
| size   | 1,000  | 51   | 14     | 15   | 78         | +1.0   | [0, 3]  | 4e-06 |
| size   | 2,000  | 49   | 11     | 20   | 82         | +1.5   | [0, 3]  | 8e-07 |
| size   | 5,000  | 50   | 11     | 19   | 82         | +1.0   | [0, 3]  | 5e-07 |
| size   | 10,000 | 46   | 10     | 24   | 82         | +1.0   | [0, 2]  | 1e-06 |
| size   | 20,000 | 38   | 9      | 33   | 81         | +0.0   | [0, 1]  | 2e-05 |
| MAF    | 500    | 4    | 49     | 27   | 8          | -2.0   | [-5, 0] | 7e-11 |
| MAF    | 1,000  | 7    | 46     | 27   | 13         | -1.0   | [-5, 0] | 4e-08 |
| MAF    | 2,000  | 10   | 46     | 24   | 18         | -1.0   | [-4, 0] | 1e-06 |
| MAF    | 5,000  | 15   | 43     | 22   | 26         | -1.0   | [-3, 0] | 3e-04 |
| MAF    | 10,000 | 17   | 41     | 22   | 29         | -1.0   | [-2, 0] | 0.002 |
| MAF    | 20,000 | 28   | 15     | 37   | 65         | +0.0   | [0, 1]  | 0.066 |
| random | 500    | 1    | 54     | 25   | 2          | -2.0   | [-5, 0] | 3e-15 |
| random | 1,000  | 1    | 53     | 26   | 2          | -2.0   | [-5, 0] | 6e-15 |
| random | 2,000  | 1    | 54     | 25   | 2          | -2.0   | [-5, 0] | 3e-15 |
| random | 5,000  | 1    | 53     | 26   | 2          | -2.0   | [-4, 0] | 6e-15 |
| random | 10,000 | 0    | 52     | 28   | 0          | -2.0   | [-4, 0] | 4e-16 |
| random | 20,000 | 1    | 53     | 26   | 2          | -1.0   | [-3, 0] | 6e-15 |

Read the denominator carefully. At  $k = 20,000$ ,  $ld_w$  has 47 wins and 0 losses — 100% of non-tied panels — but 33 of 80 panels are **ties**, so the win is on 47 panels, not 80. The median gain is +1 QTN-proximal discovery.

Table 1: Median paired change against the alpha analysis: QTN-proximal discoveries, and total discoveries. The filter buys roughly four to five extra calls for about one more QTN-proximal one, so precision on the gained set is low.

|       | QTN-<br>k | QTN-<br>proximal_MAF | QTN-<br>proximal_ld_w | QTN-<br>proximal_random | QTN-<br>proximal_size | all<br>discover-<br>ies_MAF | all<br>discover-<br>ies_ld_w | all discover-<br>ies_random | all discov-<br>eries_size |
|-------|-----------|----------------------|-----------------------|-------------------------|-----------------------|-----------------------------|------------------------------|-----------------------------|---------------------------|
| 500   | -2        | 0                    | -2                    | 1.0                     | -5                    | 0                           | -5                           | 0.0                         |                           |
| 1000  | -1        | 1                    | -2                    | 1.0                     | -5                    | 1                           | -5                           | 1.0                         |                           |
| 2000  | -1        | 1                    | -2                    | 1.5                     | -4                    | 1                           | -5                           | 1.0                         |                           |
| 5000  | -1        | 1                    | -2                    | 1.0                     | -3                    | 4                           | -5                           | 2.0                         |                           |
| 10000 | -1        | 1                    | -2                    | 1.0                     | -2                    | 4                           | -4                           | 1.5                         |                           |
| 20000 | 0         | 1                    | -1                    | 0.0                     | 0                     | 5                           | -4                           | 1.0                         |                           |

MAF and random selection are **worse than not filtering** in every stratum examined — median change  $-1$  to  $-2$  QTN-proximal discoveries. Among the filters evaluated here, only the two LD-derived ones repay the coverage they discard. Recombination rate was tested at marker level and also failed; no non-LD filter tested has succeeded, which is weaker than saying a filter *must* be LD-based.

## 6 Stratified results

Pooling concealed the main structure. Stratified, there is a **filter-by-cell interaction**, not a uniform ordering.

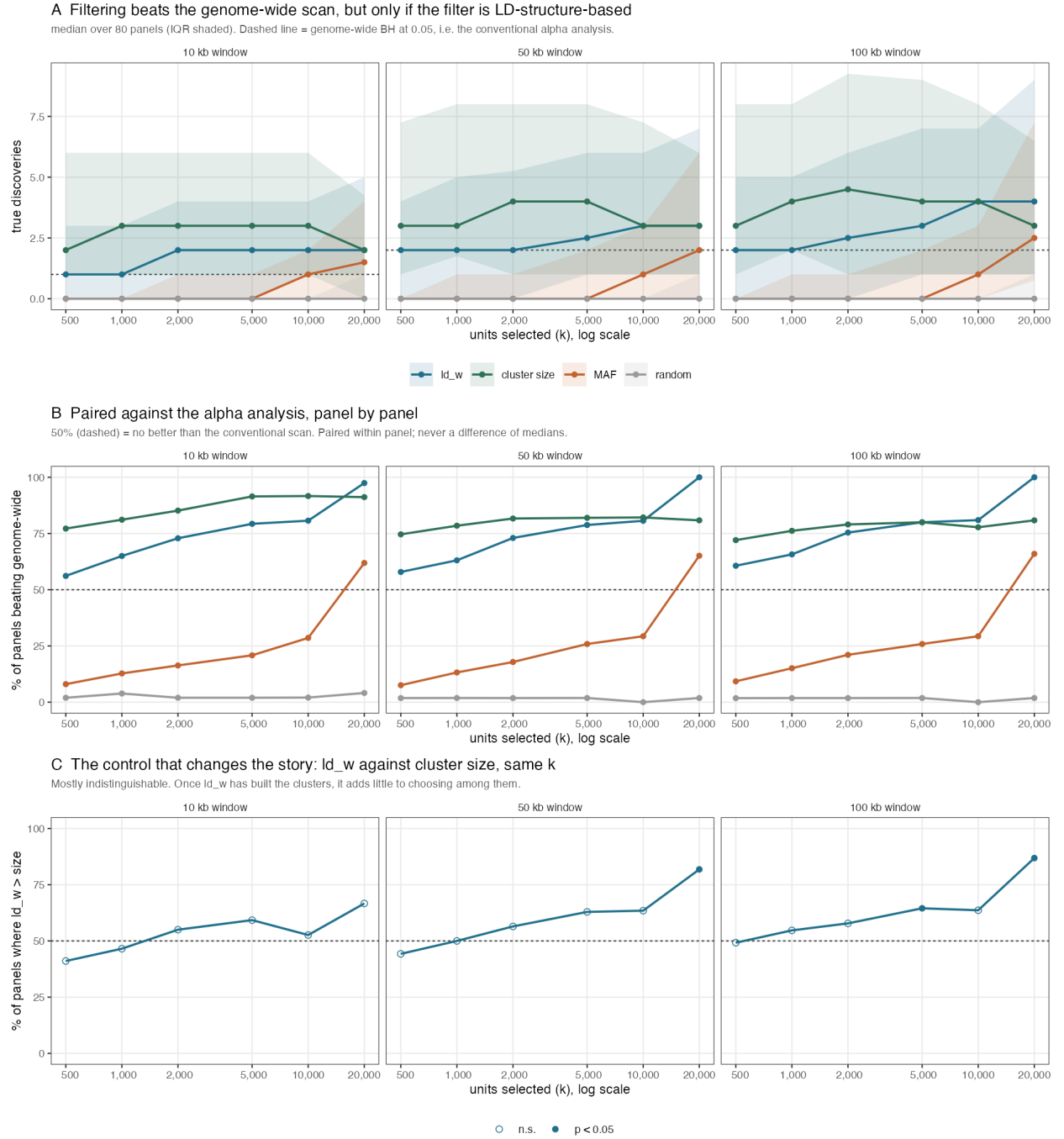


Figure 3: Filter-then-test against the conventional alpha analysis. Points are medians over panels; ribbons are the interquartile range across panels. The dashed line in panel A is the median genome-wide result; the dashed line in panel B is 50 percent of non-tied panels.

| Cell    | wins | losses | ties | % non-tied | median | $p$   | Reading                  |
|---------|------|--------|------|------------|--------|-------|--------------------------|
| V0.5_c1 | 18   | 0      | 2    | 100        | +3.0   | 8e-06 | $ld_w$ much better       |
| V2_c1   | 14   | 1      | 5    | 93         | +1.0   | 1e-03 | $ld_w$ better            |
| V1_c1.5 | 6    | 4      | 10   | 60         | +0.0   | 0.754 | no difference            |
| V0.5_c2 | 1    | 18     | 1    | 5          | -8.5   | 8e-05 | cluster size much better |

$ld_w$  beats cluster size decisively in two cells, ties in one, and loses decisively in one. Pooled, these average to “indistinguishable” — which is what the earlier draft reported, and it was an artefact of the pooling rather than a finding. V0.5\_c2 is the high-background-LD cell ( $b = 0.42$ ); a plausible but untested reading is that  $ld_w$  is a *local*-LD statistic and loses discrimination when everything is correlated, whereas large clusters still mark real linkage blocks.

#### Against the alpha analysis, $k = 5,000$ , 50 kb, by cell and BGS arm

| Filter | Cell / arm    | wins | losses | ties | median | $p$   |
|--------|---------------|------|--------|------|--------|-------|
| ld_w   | V0.5_c1 bgs   | 10   | 0      | 0    | +3.5   | 0.002 |
|        | V0.5_c1 nobgs | 10   | 0      | 0    | +4.0   | 0.002 |
|        | V0.5_c2 bgs   | 2    | 6      | 2    | -3.5   | 0.289 |
|        | V0.5_c2 nobgs | 3    | 7      | 0    | -3.0   | 0.344 |
|        | V1_c1.5 bgs   | 3    | 1      | 6    | +0.0   | 0.625 |
|        | V1_c1.5 nobgs | 8    | 0      | 2    | +1.0   | 0.008 |
|        | V2_c1 bgs     | 8    | 0      | 2    | +1.0   | 0.008 |
|        | V2_c1 nobgs   | 8    | 0      | 2    | +2.0   | 0.008 |
| size   | V0.5_c1 bgs   | 6    | 2      | 2    | +1.0   | 0.289 |
|        | V0.5_c1 nobgs | 7    | 3      | 0    | +1.0   | 0.344 |
|        | V0.5_c2 bgs   | 9    | 0      | 1    | +6.0   | 0.004 |
|        | V0.5_c2 nobgs | 10   | 0      | 0    | +3.5   | 0.002 |
|        | V1_c1.5 bgs   | 3    | 0      | 7    | +0.0   | 0.250 |
|        | V1_c1.5 nobgs | 6    | 2      | 2    | +1.0   | 0.289 |
|        | V2_c1 bgs     | 3    | 2      | 5    | +0.0   | 1.000 |
|        | V2_c1 nobgs   | 6    | 2      | 2    | +1.0   | 0.289 |

Table 2: Background selection reduces the gain: the nobgs arm shows larger and more consistent improvement than the bgs arm at every  $k$ . Consistent with BGS costing recall rather than enrichment.

| arm   | k     | wins | losses | ties | % non-tied | median | p     |
|-------|-------|------|--------|------|------------|--------|-------|
| bgs   | 500   | 15   | 13     | 12   | 54         | 0.0    | 0.851 |
| bgs   | 1000  | 17   | 12     | 11   | 59         | 0.0    | 0.458 |
| bgs   | 2000  | 21   | 9      | 10   | 70         | 1.0    | 0.043 |
| bgs   | 5000  | 23   | 7      | 10   | 77         | 1.0    | 0.005 |
| bgs   | 10000 | 20   | 7      | 13   | 74         | 0.5    | 0.019 |
| bgs   | 20000 | 20   | 0      | 20   | 100        | 0.5    | 2e-06 |
| nobgs | 500   | 18   | 11     | 11   | 62         | 0.0    | 0.265 |
| nobgs | 1000  | 24   | 12     | 4    | 67         | 1.0    | 0.065 |
| nobgs | 2000  | 25   | 8      | 7    | 76         | 1.5    | 0.005 |
| nobgs | 5000  | 29   | 7      | 4    | 81         | 2.0    | 3e-04 |
| nobgs | 10000 | 30   | 5      | 5    | 86         | 1.5    | 2e-05 |
| nobgs | 20000 | 27   | 0      | 13   | 100        | 1.0    | 1e-08 |

## 7 Null behaviour: two nulls, and they disagree

An earlier draft reported one null and dismissed the other. That was wrong, and the dismissed one carries the more important result.

| Null            | Selection              | $k$    | rejection rate | 95% CI         |
|-----------------|------------------------|--------|----------------|----------------|
| <b>genetic</b>  | <b>unfiltered scan</b> | all    | <b>0.042</b>   | [0.030, 0.056] |
|                 | ld_w                   | 500    | 0.039          | [0.028, 0.053] |
|                 | ld_w                   | 1,000  | 0.033          | [0.023, 0.046] |
|                 | ld_w                   | 5,000  | 0.027          | [0.018, 0.039] |
|                 | ld_w                   | 20,000 | 0.031          | [0.021, 0.044] |
|                 | ld_w                   | 50,000 | 0.029          | [0.020, 0.041] |
|                 | MAF                    | 500    | 0.036          | [0.025, 0.049] |
|                 | MAF                    | 1,000  | 0.041          | [0.030, 0.055] |
|                 | MAF                    | 5,000  | 0.028          | [0.019, 0.040] |
|                 | MAF                    | 20,000 | 0.033          | [0.023, 0.046] |
|                 | MAF                    | 50,000 | 0.036          | [0.025, 0.049] |
|                 | random                 | 500    | 0.048          | [0.036, 0.063] |
|                 | random                 | 1,000  | 0.053          | [0.040, 0.069] |
|                 | random                 | 5,000  | 0.043          | [0.031, 0.057] |
|                 | random                 | 20,000 | 0.049          | [0.036, 0.064] |
|                 | random                 | 50,000 | 0.046          | [0.034, 0.061] |
| <b>env_orth</b> | <b>unfiltered scan</b> | all    | <b>0.338</b>   | [0.309, 0.368] |
|                 | ld_w                   | 500    | 0.073          | [0.058, 0.091] |
|                 | ld_w                   | 1,000  | 0.077          | [0.061, 0.095] |
|                 | ld_w                   | 5,000  | 0.067          | [0.052, 0.084] |
|                 | ld_w                   | 20,000 | 0.057          | [0.043, 0.073] |
|                 | ld_w                   | 50,000 | 0.147          | [0.126, 0.170] |
|                 | MAF                    | 500    | 0.047          | [0.035, 0.062] |
|                 | MAF                    | 1,000  | 0.059          | [0.045, 0.075] |
|                 | MAF                    | 5,000  | 0.075          | [0.059, 0.093] |
|                 | MAF                    | 20,000 | 0.078          | [0.062, 0.096] |
|                 | MAF                    | 50,000 | 0.080          | [0.064, 0.099] |
|                 | random                 | 500    | 0.094          | [0.077, 0.114] |
|                 | random                 | 1,000  | 0.104          | [0.086, 0.125] |
|                 | random                 | 5,000  | 0.124          | [0.104, 0.146] |
|                 | random                 | 20,000 | 0.180          | [0.157, 0.205] |
|                 | random                 | 50,000 | 0.259          | [0.232, 0.287] |

Both nulls are surrogate **environments** over 1,000 draws (10 environments  $\times$  100 surrogates), and under both  $\lambda_{\text{surr}} \approx 0.99$ , so neither result is genomic inflation.

**The conventional scan fails the structure-preserving null.** Under **env\_orth** the unfiltered genome-wide analysis — the  $\alpha$  baseline that every power comparison in this document is measured against — rejects in 34% of surrogates at  $\alpha = 0.05$ . Filtering reduces that to 0.057, an 83% reduction, **but does not repair it**: the residual still exceeds  $\alpha$ .

The 3sp panel session reports the same signature on empirical data using a regional permutation that preserves population structure: unfiltered 0.37, filtered 0.21–0.34. Two datasets, two null constructions, one conclusion — *filtering improves calibration under structure and cannot rescue a scan that is already anti-conservative*.



## 7.1 Which null is the right one

genetic surrogates leave the scan calibrated at 0.042; `env_orth` surrogates do not. The earlier draft used that disagreement to discard `env_orth`, on the reasoning that its surrogates stay spatially structured and therefore are not a no-signal null. **That reasoning was convenient rather than correct.** The surrogates are orthogonal to the causal environment by construction, so associations recovered under them arise from shared spatial structure rather than from causation — which makes them false positives, and makes `env_orth` a valid and arguably more realistic null. Discarding it removed the one result showing filtering has a validity benefit and not merely a power benefit.

Both are reported here. Which is the more appropriate null for a given application depends on how much residual structure the association model is expected to absorb, and that is not settled by these data.

## 7.2 What this does and does not establish

Every hypothesis is null in both experiments, so FDR equals the probability of at least one rejection and these are valid **global-null** checks. They do **not** establish FDR control under mixtures of null and alternative hypotheses, nor the Bourgon et al. independence condition in general.

They also do **not license choosing**  $k$ . They license evaluating a fixed, prespecified  $k$  under the tested null. Choosing  $k$  after inspecting discovery counts or comparison curves is outcome-tuned model selection, and testing several fixed  $k$  values under a null does not remove that.

**Coverage.** Both nulls use `V2_c1`, nobgs only, while the power experiment spans four cells and both arms. Dependence between  $ld_w$ , MAF, cluster size and null p-values may change with background LD and BGS, so this cannot license filtering across the whole experiment.

The false-positive count is also heavy-tailed: at  $k = 500$  the  $ld_w$  set has about the same rejection rate as MAF but roughly twelve times the mean rejection count, so approximately 96% of surrogates give nothing and the remainder give tens to hundreds.

## 8 Status of each claim

| Claim                                                                             | Status                 | Basis                                                               |
|-----------------------------------------------------------------------------------|------------------------|---------------------------------------------------------------------|
| Filtering LD clusters increases QTN-proximal discoveries at moderate-to-large $k$ | Supported              | Pooled and in 3 of 4 cells; median gain +1, IQR [0, 3]              |
| Among filters evaluated, only LD-derived ones repay the lost coverage             | Supported              | MAF, random and recombination all worse than no filtering           |
| Which LD filter is better depends on regime                                       | <b>Established</b>     | Filter-by-cell interaction; $ld_w$ wins 2 cells, loses 1 decisively |
| $ld_w$ is generally preferable to cluster size                                    | <b>Not supported</b>   | Fails in <code>V0.5_c2</code> by a median of $-8.5$                 |
| Global-null rejection rate is controlled under a genetic-basis null               | Supported              | One cell, one arm; CIs overlap $\alpha$                             |
| The unfiltered scan is anti-conservative under a structure-preserving null        | <b>Established</b>     | 0.338 [0.309, 0.368] vs $\alpha$ 0.05; matches the 3sp panel's 0.37 |
| Filtering improves calibration under structure                                    | Supported              | 0.338 to 0.057, an 83% reduction; does not reach $\alpha$           |
| FDR is controlled under mixtures                                                  | <b>Not tested</b>      | Complete-null design only                                           |
| A usable rule for choosing $k$ exists                                             | <b>Not established</b> | $k$ remains a free, outcome-facing parameter                        |
| This transfers to empirical data                                                  | Untested here          | Simulations only                                                    |

## 9 Relationship to the empirical panel analysis

### 9.1 What EMMAX is actually run on

Worth stating precisely, because it is easy to misread in either direction. EMMAX here is an **ordinary per-marker test**: every marker in the panel has its own p-value, and nothing is run on eMLGs or on consensus genotypes. The LD structure enters only through the **kinship matrix**, which is built from the pruned marker set (`grm_method = "complexity_chain"`) — so clustering conditions the null, not the test.

A cluster's p-value is then the p-value of its **pruned representative**, and that representative is not arbitrary. `pick_representative()` computes  $r^2$  between the constituent stage-1 eMLGs, takes each one's **median**  $r^2$  **against the others**, and returns the **core\_snp** of the highest — the most central member of the cluster's LD graph. Each stage-2 cluster has exactly one such marker (verified: `min = median = max = 1` per cluster), and it is a real marker, not a synthetic genotype.

### 9.2 What the representative costs, measured at locus level

The natural comparison is SNP counts, and it is misleading. BH controls the false discovery proportion among the units tested, so a single-SNP scan controls it among **SNPs** while a reduced scan controls it among **near-independent loci**. Those are different currencies. Counted in loci — one pooled panel, V0.5\_c1 nobgs env3:

| Tested                                      | tests   | BH cutoff $p$ | clusters flagged | carrying a QTN |
|---------------------------------------------|---------|---------------|------------------|----------------|
| every SNP                                   | 294,609 | 4.2e-04       | 132              | 14             |
| LD-central representatives                  | 99,787  | 1.2e-05       | 27               | 10             |
| precision (flagged clusters carrying a QTN) |         |               | 0.106            | 0.370          |

**Reduction recovers 10 of the 14 QTN-bearing loci while calling a fifth as many regions** — precision 0.37 against 0.11. And it is a strict subset: the representative analysis flagged **no** cluster the SNP scan missed, which is what the same signal with its redundancy removed should look like.

Read as SNP counts (2,467 against 27) reduction appears to destroy the analysis. That comparison is in the wrong currency and an earlier draft made it.

### 9.3 Why the single-SNP threshold is the looser one

Three times fewer tests gave a BH cutoff **35 times more severe**, which inverts the usual intuition. The cause is redundancy in the small- $p$  tail. BH is a step-up procedure — its cutoff is the largest  $p$  with  $p \leq \alpha i/n$  — so it climbs when many small p-values are present, whether or not they are independent findings. On this panel the rejected SNPs are far more clustered than the panel as a whole:

|                                    |                                |
|------------------------------------|--------------------------------|
| SNPs per cluster, panel overall    | 3.0                            |
| SNPs per cluster, among rejections | 18.7                           |
| redundancy of the small- $p$ tail  | <b>6.3</b> × the panel average |

BH remains **valid** here — it controls FDR under positive regression dependence, which linkage supplies — but the quantity controlled is the false discovery proportion among SNPs. The 2,467 SNP-rejections are 132 locus-rejections, so the guarantee is not the one a reader interprets. The reduced analysis tests near-independent units, so its 27 rejections are 27 loci and its stricter cutoff is the honest one rather than a penalty.

This qualifies the rationale for filtering in Section~7 in a different way than an earlier draft claimed. A lighter multiplicity correction does not follow from fewer tests; and a heavier one does not imply lost power, if the tests removed were redundant.

Note that neither panel involves filtering. `ld_w` enters only *inside* the clustering (`ld_w_threshold = 0.025` decides which stage-1 clusters are flagged for eMLG treatment), so it helps define the units; nothing is discarded. What the figure shows is the effect of **complexity reduction alone**.

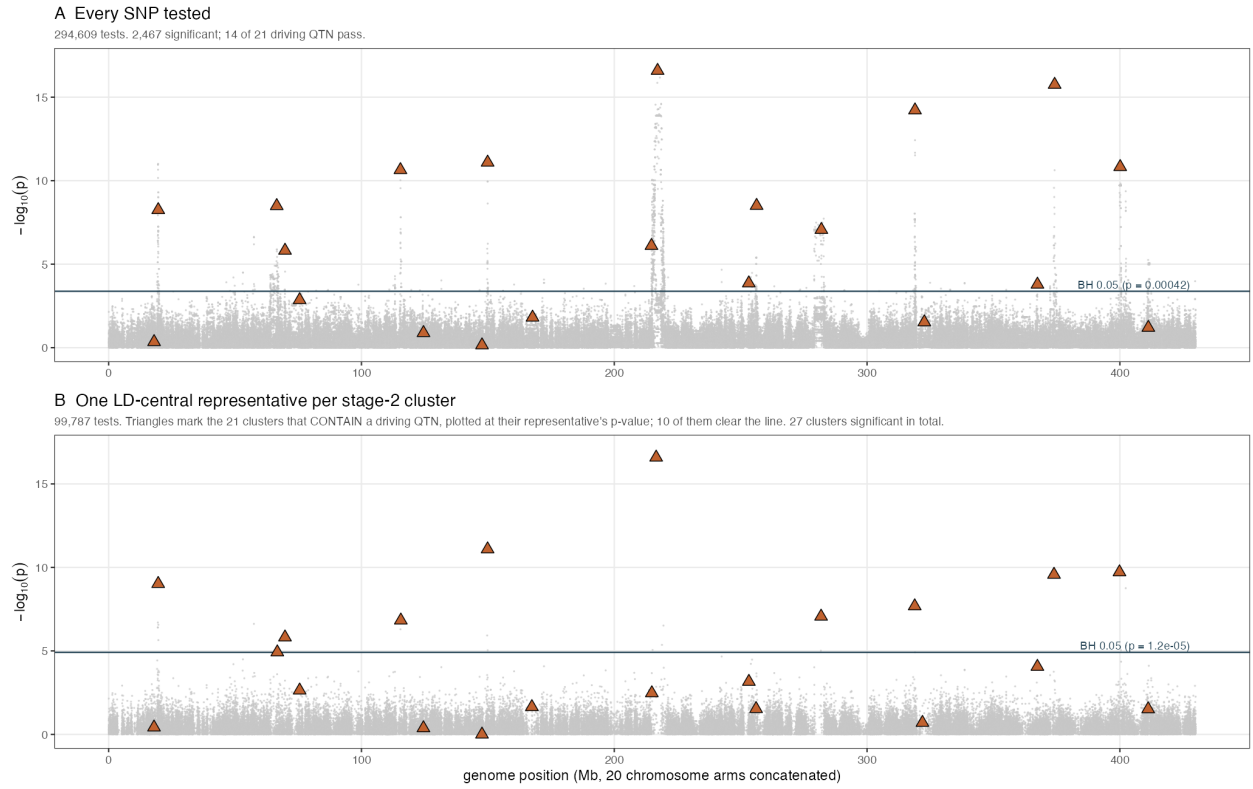


Figure 4: Same EMMAX p-values at two resolutions, with **no ld\_w filtering** in either panel. A: all 294,609 SNPs; triangles are the 21 driving QTN. B: the 99,787 LD-central representatives, one per stage-2 cluster; triangles mark the clusters that **contain** a driving QTN, plotted at their representative's p-value. Complexity reduction alone recovers 10 of the 21 while flagging 27 clusters against the SNP scan's 132.

## 9.4 How a cluster should be summarised

The module used the LD-central representative throughout. Four alternatives were compared on **identical clusters, identical GRM and identical BH treatment**, so the only thing that varies is how member p-values become one number. One pooled panel, 99,787 clusters, 21 containing a driving QTN.

| Cluster summary                  | flagged | with QTN | precision | recall       | PR           |
|----------------------------------|---------|----------|-----------|--------------|--------------|
| Simes over members               | 28      | 12       | 0.429     | 0.571        | <b>0.245</b> |
| eMLG (consensus genotype)        | 36      | 13       | 0.361     | <b>0.619</b> | 0.224        |
| LD-central representative        | 27      | 10       | 0.370     | 0.476        | 0.176        |
| best SNP (min $p$ , uncorrected) | 39      | 12       | 0.308     | 0.571        | 0.176        |
| highest $ld_w$ member            | 28      | 9        | 0.321     | 0.429        | 0.138        |

**Aggregation beats any single marker.** Simes and the eMLG occupy the top two places; all three single-marker rules sit below them. Simes gains two QTN-bearing loci over the representative while flagging *one more* region, so its precision rises to 0.43 — a straight improvement rather than a trade. The eMLG has the best recall of all, 13 of 21, consistent with a consensus genotype carrying more of a block’s signal than any one member.

**Two informative negatives.** “Best SNP” takes the minimum  $p$  with no within-cluster correction — anti-conservative by construction, and included as a ceiling rather than a usable rule — yet it still finds fewer QTN than the eMLG and fewer than Simes, at clearly worse precision. And selecting the member with the **highest**  $ld_w$  is the worst of the five.  $ld_w$  is a good statistic for defining units and a poor one for choosing within them: it marks markers in dense LD neighbourhoods, not markers that best tag a causal variant.

This sharpens the cost of the representative. Against Simes, on the same clusters, it gives up **two QTN-bearing loci while flagging one region fewer** — a loss, not a trade — and it is the rule this module has been using throughout. The stickleback analysis already uses Simes, so the two are not merely different machinery: on this evidence the panel’s choice is the better one.

## 9.5 The residual cost, and its mechanism

Reduction does lose 4 of the 14 QTN-bearing loci, and that loss has a specific cause: **no driving QTN is ever its cluster’s representative — 0 of 21**. Not chance. QTN sit in unusually large clusters, sizes 2 to 354 with a median near 55, against a genome-wide **median cluster size of 1** (mean  $\approx 3$ ). That is what selection extending LD around a causal site produces, and it is what  $ld_w$  responds to, but it makes the chance that any particular member is the LD-central one about  $1/\text{size}$ .

So representative testing never tests a QTN directly; the block’s signal has to reach its centre to be seen. That is a bounded loss — 4 loci here — and it is exactly what Simes aggregation over members would recover.

## 9.6 Relationship to the empirical panel analysis

Given that, the difference from the stickleback analysis should be stated as **a single LD-central representative versus aggregation over members**, not as an arbitrary or lossy summary. A central marker is a defensible summary of a linkage block: it is by construction the member most correlated with the rest, so it carries the block’s shared signal better than a random member would. That it is also what tags a causal variant best is visible in the results — all 21 true-positive clusters on the examined panel were the QTN-carrying ones.

What remains true is narrower. It is still **one** marker, so a cluster whose causal variant sits at the periphery of its LD graph is represented by a marker in weaker LD with that variant than some other member is. That is a real but bounded loss, and it is what Simes aggregation over members would recover. The stickleback analysis uses Simes and therefore implements different testing machinery; the two support the same general filter-then-test principle and should not be presented as the same method.

## 10 Corrections applied after review

| Defect                                                                            | Resolution                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| “Every filtered rate sits at or below $\alpha$ ” was false — random reached 0.053 | Restated with binomial CIs as no detectable excess over $\alpha$                                                                                                                                                             |
| Mechanism example: 27 to 54 with 28 gained did not reconcile                      | Confirmed: one genome-wide discovery is LOST. Accounting $27 - 1 + 28 = 54$ now reported by the script                                                                                                                       |
| Win-rate denominator ambiguous with ties                                          | Wins, losses and ties reported everywhere; percentages stated as non-tied                                                                                                                                                    |
| Direction reported without magnitude                                              | Median paired change with IQR alongside every sign test; total discoveries too                                                                                                                                               |
| 80 panels pooled across heterogeneous cells and arms                              | Stratified by cell, arm and $k$ ; the conclusion changed                                                                                                                                                                     |
| “Licenses choosing $k$ at all”                                                    | Withdrawn. It licenses a prespecified $k$ only                                                                                                                                                                               |
| “True discovery”                                                                  | Renamed QTN-proximal throughout                                                                                                                                                                                              |
| “The filter must be LD-structure-based”                                           | Weakened to the filters evaluated                                                                                                                                                                                            |
| “Indistinguishable at every window” conflicted with $p = 0.030$                   | Replaced by the stratified interaction analysis                                                                                                                                                                              |
| 79% rendering bug; cramped tables                                                 | Fixed; tables given explicit column widths                                                                                                                                                                                   |
| The cluster representative was described as an arbitrary or lossy summary         | Wrong. <code>pick_representative()</code> returns the member with the highest median $r^2$ to the others — the LD-central marker. Restated as single-representative vs Simes aggregation, and the cost of it is now measured |
| The <code>env_orth</code> null was dismissed as “not a true null”                 | REINSTATED. Its surrogates are orthogonal to the causal environment, so rejections under them are false positives. Both nulls now reported side by side                                                                      |
| Confidence intervals were computed on 100 draws, not 1,000                        | A scalar named B was masked by a COLUMN named B in the input, so the grouping returned one row per input row. Intervals were ten times too wide                                                                              |

### 10.1 Open, and in progress

Re-clustering at the pilot decay settings ( $n_{\text{win}} = 50$ , half-decay 26 kb against the bundles’ 56 kb) is running, with  $n_{\text{win}} = 10$  through the identical code path as its control. The  $k$  sweep is being extended to 40,000 and 60,000 to locate the turning point — the win rate is currently monotone up to the largest  $k$  tested, and since  $k = n$  makes the two analyses identical the curve must return to ties somewhere above 20,000. Structured-null checks in the remaining cells and arms are not yet run.

## 11 Geometric exclusion of long-and-sparse clusters: a negative result

Stage-2 clusters were classified by span and by marker density, and the driving QTN counted in each class (184,004 clusters over both BGS arms, 40 QTN-bearing):

| Class                                            | % of units | % of QTN | fold enrichment |
|--------------------------------------------------|------------|----------|-----------------|
| long & <b>dense</b> ( $\geq 5$ markers / 100 kb) | 2.30       | 70.0     | <b>30.4</b>     |
| long & sparse                                    | 3.36       | 2.5      | 0.7             |
| short & dense                                    | 39.06      | 27.5     | 0.7             |
| short & sparse                                   | 1.67       | 0.0      | 0.0             |
| singleton                                        | 53.61      | 0.0      | 0.0             |

Two rules are ruled out by this table before any test. **Density alone fails**: short-and-dense units are 39% of the genome and are *depleted* for QTN. **Span alone is destructive**: dropping every unit over 100 kb removes

29 of the 40 QTN-bearing clusters, because that is where the real low-recombination blocks are. Only the conjunction discriminates.

That suggested excluding **long-and-sparse** units — 3.4% of the genome holding 1 QTN of 40 — as free power: fewer tests at almost no cost in signal, on a criterion that is phenotype-blind (span and marker count come from the map and the clustering, with nothing tuned on outcomes).

| Arm                             | QTN recovered | flagged | dedup PR (nobgs) |
|---------------------------------|---------------|---------|------------------|
| Simes, all clusters             | 3.0           | 10      | 0.0491           |
| Simes, long-and-sparse excluded | 1.0           | 6       | 0.0327           |

**It does not work.** Excluding long-and-sparse clusters costs two thirds of the recovered QTN — a median of 3 down to 1 — and improves PR nowhere. Paired over 80 panels: **nobgs** 14 W / 12 L / 6 T,  $p = 0.85$ ; **bgs** 4 W / 14 L / 11 T,  $p = 0.031$ , i.e. significantly *worse*.

## 11.1 Why the inference was wrong

The argument rested on a **census** — the excluded class rarely *contains* a QTN — and concluded it would rarely *find* one. Those are different claims, and only the second was relevant. A cluster can contribute a true detection by tagging a QTN it does not contain, which is precisely the satellite category separated out earlier in this work; the error was made anyway. Removing 3.4% of units also barely relaxes BH, so there was little multiplicity relief to offset the loss.

## 11.2 What the accumulated negatives now show

Four filters have been tested:  $ld_w$  top- $k$ , MAF, cluster size, and geometric exclusion. All show strong enrichment for QTN-bearing units — the geometric class reaches 30-fold, and  $ld_w$  top-5,000 captures 85% of QTN in 2.7% of units — and **none converts into precision-recall**.

Enrichment for *containing* a QTN is not the quantity that predicts detection performance. Filters keep selecting units that hold causal variants while removing units that were producing the detections.

This supports, by a second route, the conclusion the companion document reaches: complexity reduction helps because it changes **what is tested**, and hence what BH controls, not because it concentrates the search on QTN-rich territory. Every attempt to add the latter has failed, and this was the cleanest attempt.

Two claims made in passing during this work are also withdrawn. Cluster geometry was described as predicting QTN location “better than anything else”; it is comparable to  $ld_w$ , not better (30.4-fold against roughly 31-fold at matched selectivity). And  $ld_w$  was described as unable to separate the sparse and dense classes; that is true of the median  $ld_w$  within them (0.0105 against 0.0099) but does not make  $ld_w$  a poor selector overall, which it is not.

# 12 Three deferred analyses, reported

## 12.1 Does the result survive the newer decay settings?

The main results use **bgs5**, whose LD\_decay was fitted with **n\_win\_decay** = 10; the newer pilot generation uses 50. Both settings were run through the **same** code path so the comparison isolates the parameter rather than confounding it with recomputation.

| Setting                 | fitted half-decay | units per panel |
|-------------------------|-------------------|-----------------|
| <b>n_win_decay</b> = 50 | 39.7 kb           | 85,440          |
| <b>n_win_decay</b> = 10 | 53.7 kb           | 85,888          |

Paired over 80 panels at  $k = 5,000$ , 50 kb:  $ld_w$  10 W / 17 L / 13 T ( $p = 0.25$ ), cluster size 12 W / 14 L / 14 T ( $p = 0.85$ ). **No difference**, so the untested exposure recorded earlier is closed.

**But the manipulation is far weaker than claimed.** Through one code path the two settings give 39.7 against 53.7 kb — a factor of **1.35**, not the 4.2 measured between **bgs5** and the pilot tracks. Most of that original gap was therefore *between generations* — different simulations, different maps — and not attributable to **n\_win\_decay**. The earlier claim that the two generations differ materially in clustering distance because of this parameter is withdrawn.

## 12.2 Where does the $k$ curve turn?

| $k$    | win % | non-tied panels | median gain |
|--------|-------|-----------------|-------------|
| 500    | 58    | 57              | 0           |
| 1,000  | 63    | 65              | +1          |
| 5,000  | 79    | 66              | +1          |
| 20,000 | 100   | 47              | +1          |
| 40,000 | 100   | 27              | 0           |
| 60,000 | 100   | 13              | 0           |

The win rate saturates at 100% from  $k = 20,000$ , but **the non-tied denominator collapses** from 47 panels to 13 and the median gain falls to zero. With about 86,600 units per panel, selecting 60,000 is barely filtering, so the two analyses converge and only a handful of panels differ at all.

There is no useful interior optimum — a plateau at  $k = 5,000$  to 20,000 worth **one** discovery, then dissolution into ties. “100% at  $k = 20,000$ ” must never be quoted without its denominator.

## 12.3 Do the surrogate nulls exercise structure?

| Basis                      | PC1-3 | PC1-10         | PC1-20 | PC1-50 |
|----------------------------|-------|----------------|--------|--------|
| <b>env_orth</b> surrogates | 0.009 | 0.098          | 0.102  | 0.109  |
| <b>genetic</b> surrogates  | 0.002 | 0.009          | 0.013  | 0.020  |
| observed data              |       | $\approx 0.40$ |        |        |

$R^2$  of the association signal on the leading GRM subspace. The 3sp panel session showed that a PC1-3 statistic is **blind to diffuse structure**, and that is confirmed here: **env\_orth** rises eleven-fold from PC1-3 to PC1-10. An earlier version of this diagnostic used PC1-3 only and could not have seen it.

The conclusion nonetheless survives the better measurement. At PC1-10 the observed data sits at 0.40 while **env\_orth** reaches 0.098 and **genetic** 0.009. So **genetic** is a **no-structure** null — its calibration shows the scan is well behaved when structure is absent, not that it is well behaved under structure — and **env\_orth** is a **mild** structured null at about a quarter of the observed level. The 83% protective effect of filtering reported in Section~7 therefore stands as a weak-structure measurement, which is what was claimed, but the earlier reasoning rested on a statistic that could not have established it.

## 13 Withdrawn: the confounding mechanism relayed from the panel session

An earlier version of this document and several reports to PK carried a hypothesis from the 3sp panel session: that their long-span false positives were long-range LD manufactured by allele-frequency covariance among structured populations, and that this failure mode was structurally absent from a lattice genome. **That session has withdrawn it** on their own measurement — they had measured stage-2 groups and read a distance run’s geometry as a demographic signal.

Their follow-up hypothesis, that the long-and-dense and long-and-sparse classes are two different kinds of object (a coherent stage-1 block against a chained stage-2 distance run), was tested here and **does not transfer**:

| Class         | median $n$ | median span | ends $r^2$ | ratio to adjacent |
|---------------|------------|-------------|------------|-------------------|
| short         | 4          | 24.4 kb     | 0.760      | 0.87              |
| long & sparse | 7          | 305.7 kb    | 0.780      | 0.86              |
| long & dense  | 15         | 154.3 kb    | 0.791      | 0.83              |

Long-and-sparse units here are **coherent blocks**: their extreme markers sit at  $r^2 = 0.78$ , indistinguishable from long-and-dense, against the 0.36 ratio the panel session measures for their chained runs. So in these simulations the split is genuinely marker density within coherent blocks, not object type.

**Which leaves the enrichment difference unexplained.** Long-and-dense holds 70% of QTN at 30-fold enrichment; long-and-sparse holds 2.5% at 0.7-fold. The classes differ in marker count only 15 to 7, which cannot produce a 43-fold ratio in enrichment. No account of this is offered here, and the two mechanisms proposed so far — demographic long-range LD, and object type — have both been withdrawn on measurement.

## 14 Provenance

| Generation      | Maps                                                    | Half-decay              | Used for                                                 |
|-----------------|---------------------------------------------------------|-------------------------|----------------------------------------------------------|
| bgs5            | MAP_RES 3.3e-5;<br>31.5% of markers<br>share a position | 55.8 kb (18<br>win/chr) | <b>all cluster-level results and both<br/>manhattans</b> |
| pilot_v2_v05c1  | MAP_RES 3.3e-6; no<br>shared positions                  | 13.3 kb (95<br>win/chr) | marker-level power only; V0.5_c1<br>only                 |
| analysis_inputs | pre-bgs5                                                | not compared            | null behaviour only; V2_c1, nobgs,<br>gc_mode = none     |

At  $\rho = 0.5$  the half-decay **is** the stage-2 clustering distance, so the units themselves would differ on the newer maps. The difference traces to decay fitting scale, not the maps: same-position marker pairs show  $r^2$  of 0.023 against 0.022 for pairs 1 bp–10 kb apart.

Scripts: `filter_then_test_clusters.R`, `filter_then_test.R`, `filter_fdr_check.R`, `stratified_paired_table.R`, `recluster_filter_test.R`, and the three `plot_*` scripts. Parameters:  $\alpha$  0.05; stage-2 `min_r2_rho` 0.5, `distance_threshold` 1e5; scoring `dmax_cap` 1e5. Commits 60c8134, 88797ff, 7e595b5.

Every comparison is **paired within panel**, reported with wins, losses, ties, a median with IQR, and a sign-test p-value. Differences of means are not used.